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Self-balancing unmanned bicycle based on handlebar steering control

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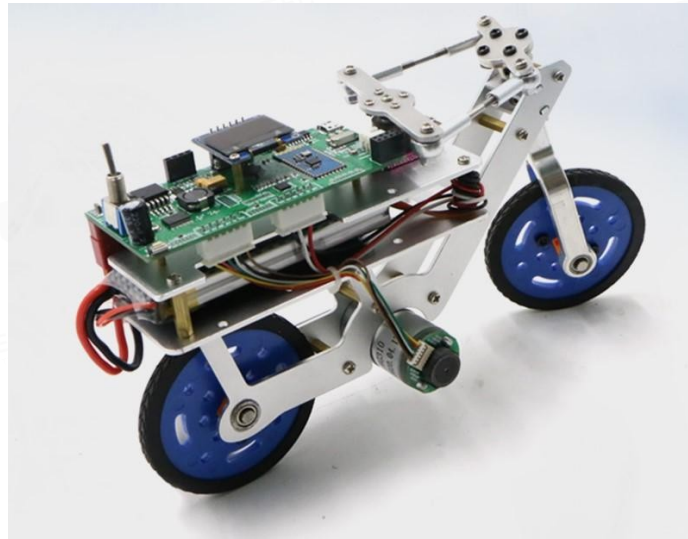
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1. kinetic model

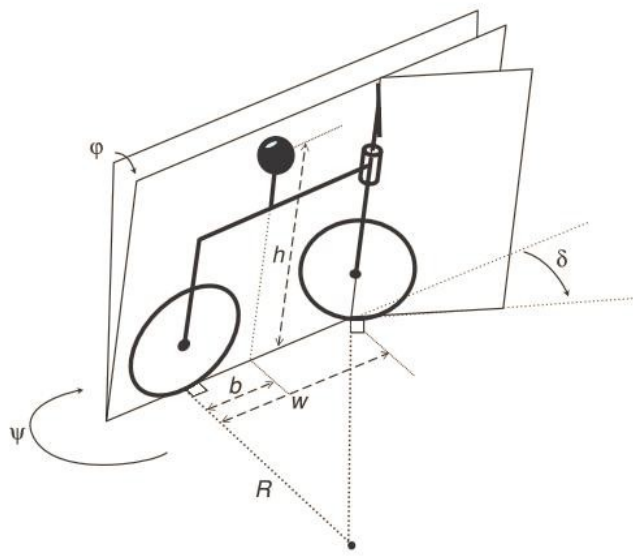
As an unmanned two-wheeled intelligent transportation vehicle, the unmanned bicycle has the characteristics of flexible movement, high energy utilization, and green environmental protection, which makes it has unique advantages and application prospects in the fields of security patrol and traffic congestion avoidance. However, on the other hand, it is a typical representative of static instability and dynamic stabilizability. In



short, it means that the bicycle cannot continuously maintain a state of equilibrium without exerting any control. For this reason, in this paper, based on the establishment of the dynamic model of the bicycle, the controller is designed to make the bicycle continuously maintain the equilibrium state.

Figure 1-1 Physical drawing of self-balancing unmanned bicycle

For ease of analysis, it is useful to simplify the bicycle model.

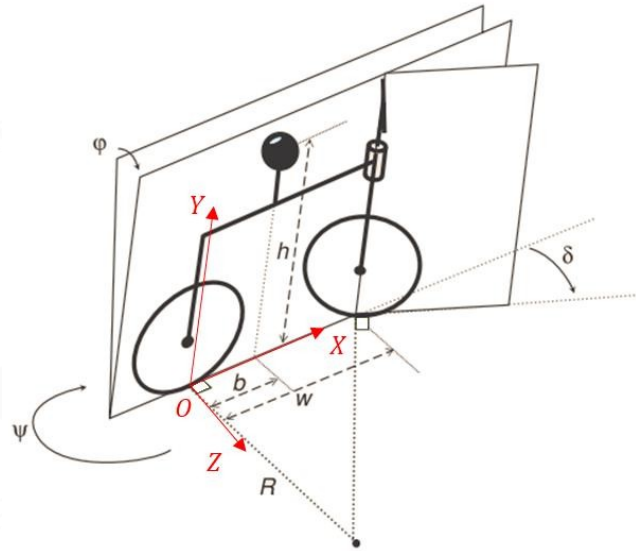


Assuming that the bicycle is moving on a horizontal plane, consider that the front fork pivot is perpendicular to the ground. The simplified model is shown in Figure 1-2.

Figure 1-2 Bicycle Model^[1]

Let the mass of the body be m , the height of the center of mass be h , the inclination of the body be ϕ , the fork angle of rotation be δ , and the front and rear wheels be

The distance between the point of contact with the ground is w , where b is the distance between the point of impact of the rear wheel and the



projection of the center of gravity on the ground is R . Establish the following coordinate system (O - XYZ) as shown below.

Fig. 1-3 Establishment of Random Coordinate System

Since the fork turning angle is relatively small, it can be assumed that the turning radius of the center of mass motion is approximately equal to the distance from the rear wheel to the center of the circle. The following relationship is satisfied:

$$\delta = \tan \delta = \frac{w}{R} \quad (1)$$

assume (office)

$$w = R\delta \quad (2)$$

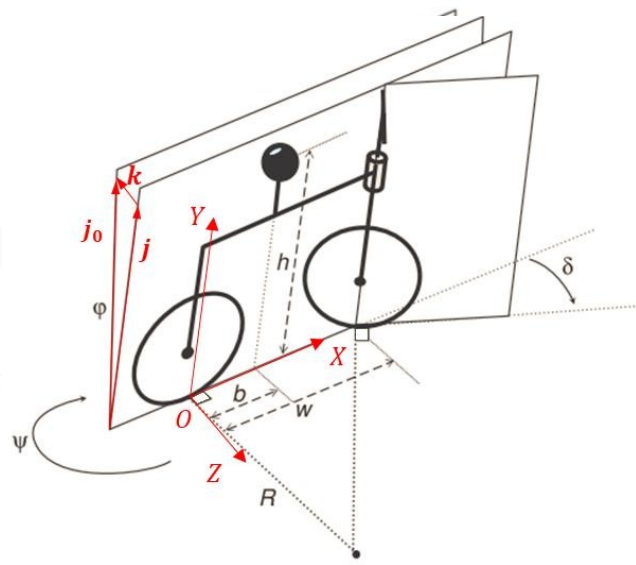


Fig. 1-4 Relationship between the basis vectors of vertical axes and the basis vectors of each axis

Let the basis vectors (here the vectors are shown in bold blue) of each axis of the coordinate system be $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$, and the center of mass of the

The position (relative to the origin O of the follower coordinate system, i.e., the point of impact of the rear wheel) can be expressed as $= +h$.

The angular velocity of the vehicle body consists of two parts: the angular velocity of rotation about the horizontal axis (X-axis) and the angular velocity of rotation about the vertical axis (not the Y-axis), where the magnitude of the angular velocity of rotation about the horizontal axis is and the magnitude of the angular velocity of rotation about the vertical axis is . (Figure 1-3 shows how the basis vectors of the vertical axis are in terms of the basis vectors of each axis.

This is a simple illustration. As a simple illustration, $= \cos - \sin$, and because the inclination of the body is small, $= -$. In fact, analogous to circular motion, is obtained by rotating by the angle , and - are the displacements. Obviously, when $\rightarrow 0$, the direction of the displacement (or the direction of the velocity) is the direction of the tangent to the point, i.e., -, and the magnitude is close to the arc length, i.e., . So $= -$, i.e. $= -$)

Well, there's

$$\omega = \dot{\varphi}i - \dot{\psi}(j - \varphi k) \quad (3)$$

included among these

$$\dot{\psi} = \frac{v}{R} \quad (4)$$

Then the velocity of the center of mass with respect to the origin O of the follower coordinate system is

$$\omega \times r = -h\dot{\psi}\varphi i + b\dot{\psi}\varphi j + (h\dot{\varphi} + b\dot{\psi})k \quad (5)$$

This gives the actual velocity of the center of mass as

$$\dot{r} = v i + \omega \times r = (v - h\dot{\psi}\varphi)i + b\dot{\psi}\varphi j + (h\dot{\varphi} + b\dot{\psi})k \quad (6)$$

From equations (2) and (4)

$$\dot{\psi} = \frac{v}{w} \delta \quad (7)$$

Also because $\delta \rightarrow 0$, so $\dot{\psi} \rightarrow 0$

Therefore (only first order small quantities are considered in the calculation)

$$\dot{r} = v i + (h\dot{\varphi} + b\dot{\psi})k \quad (8)$$

Consider a bicycle moving at a constant rate on a horizontal surface with

the following constraints

$$\begin{cases} \ddot{\mathbf{r}} \cdot \mathbf{i} = 0 \\ \ddot{\mathbf{r}} \cdot \mathbf{j} = 0 \end{cases} \quad (9)$$

From equations 80 and 90, the acceleration of the bicycle is (the base vector in the follower coordinate system is not a constant vector)

$$\ddot{\mathbf{r}} = (h\ddot{\phi} + b\ddot{\psi} + v\dot{\psi})\mathbf{k} \quad (10)$$

Assuming that all the mass of the bicycle is concentrated in the center of mass, it is possible to write the body in terms of the gravitational force and the centrifugal force including the

惯性力的作用下对 X 轴的动量矩定理^[2]

$$[\mathbf{r} \times (m\mathbf{g} - m\ddot{\mathbf{r}})] \cdot \mathbf{i} = 0 \quad (11)$$

The dynamic equations for the rotation of the car body around the horizontal axis can be obtained from the coupling equations(7)and(11)as follows

$$\ddot{\phi} = \frac{g}{h} \phi - \frac{v^2}{hw} \delta - \frac{bv}{hw} \dot{\delta} \quad (12)$$

Therefore, the

equation of state

of the system is

$$\begin{pmatrix} \dot{\phi} \\ \ddot{\phi} \\ \dot{\delta} \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 \\ \frac{g}{h} & 0 & -\frac{v^2}{hw} \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} \phi \\ \dot{\phi} \\ \delta \end{pmatrix} + \begin{pmatrix} 0 \\ -\frac{bv}{hw} \\ 1 \end{pmatrix} \dot{\delta} \quad (13)$$

On the other hand, the output

equation of the system is

$$y = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \phi \\ \dot{\phi} \\ \delta \end{pmatrix} \quad (14)$$

Eq.(13) and Eq.(14) are the state space expressions of the system.

2. Design Controller

2.1 LQR Controller

LQR (Linear Quadratic Regulator), i.e. Linear Quadratic Regulator, can get the optimal control law of state linear feedback, easy to form the closed-loop optimal control, LQR optimal control can make the original system achieve better performance index by using the inexpensive cost and is easy to realize, it is a part of the more mature development of the modern control theory. The following is a brief overview of its principles and then applied to the balancing process of unmanned bicycles.

Let the state space expression for a continuous linear time-invariant system (assuming the system is fully controllable) be

$$\begin{cases} \dot{x} = Ax + Bu \\ y = Cx + Du \end{cases} \quad (15)$$

In modern control theory, the most basic control system is the full-state feedback control system (the system block diagram is shown in Fig.

2-1 shown), the goal is to design a state feedback controller $u = -$ to control the system performance.

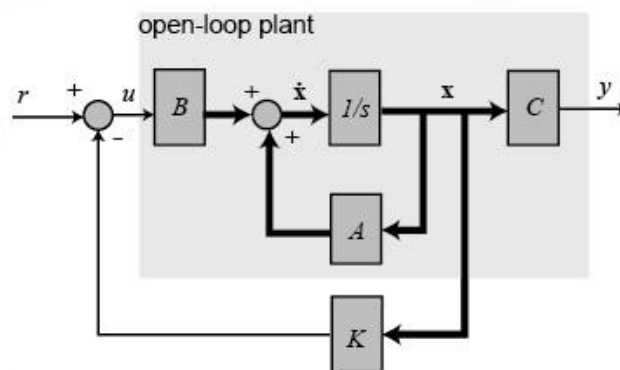


Figure 2-1 Statewide Feedback Control System

Now the problem is that such a is not unique, so how about a that is optimal? Before we introduce the cost function, let's look at this problem (from the official MATLAB tutorial): Suppose you're looking for the best way to get from your home to your office, and there are multiple transportation options available to you; you can drive a car, ride a bike, take the bus, and rent a helicopter. So, which mode of transportation is the best choice for you? There is no answer to this question per se, because we don't know what is "best". Assuming that you are only

considering the cost of time and money, fill in the following table with information about the 4 transportation options.

Table 2-1

Transportation	Time / min	Money / \$
Car	20	45
Bike	75	0

bus	30	12
Helicopter	4	2578

As you can see, if time cost is our main concern, then obviously renting a helicopter is the best option. Conversely, if we are more concerned about monetary costs, then riding a bicycle would be the best option. Of course, in real life, people tend to seek a balance between the two. For example, if you have an early meeting today, you will rightly be concerned about the time cost, but on the other hand, you are not rich enough to be concerned about the money cost as well. In that case, how do you make a choice? Here is a simple quantitative evaluation method that can create such an evaluation function $J = w_1 \cdot ie + w_2 \cdot oe$, with different weights set according to the individual's needs. If the weights are set as shown in Table 2-2, then the optimal choice is to drive to work. If the weights are set as shown in Table 2-3, then the optimal choice is to take the bus to work.

Table 2-2

Transportation	Q	Time / min	R	Money / \$	J (Cost)
Car	10	20	1	45	245
Bike	10	75	1	0	750
bus	10	30	1	12	312
Helicopter	10	4	1	2578	2618

Table 2-3

Transportation	Q	Time / min	R	Money / \$	J (Cost)
Car	5	20	5	45	325
Bike	5	75	5	0	375
bus	5	30	5	12	210
Helicopter	5	4	5	2578	12910

Similarly, for the control system, we introduce the Quadratic Cost Function (QCF)

$$J = \int_0^{+\infty} [x^T(t)Qx(t) + u^T(t)Ru(t)] dt \quad (16)$$

Where Q is a semi-positive definite state weighting matrix and R is a positive definite control weighting matrix. In engineering Q and R are symmetric matrices and often take the diagonal array. The elements on the diagonal of Q represent the weights of the corresponding error components.

x_i (The ultimate goal is to make the value of each state variable go to 0, so the value of each state variable is also called the error value.) The more

the error component is emphasized, the faster it is expected to get smaller, and accordingly, the larger its weighting coefficients are achieved. Similarly, the element i on the diagonal in represents a limit on the corresponding input component u to not be too large. The idea behind the design of the LQR controller is to design a state feedback controller

$u = -Kx$, so that the cost function is minimized $\min J$, thus achieving the goal of using the cheap cost to make the original system reach a better performance index. As to how the state feedback matrix K is calculated, only the corresponding conclusion is given here:

$$K = R^{-1}B^TP \quad (17)$$

where P is the solution of **Riccati's** equation $A^TP + PA + Q - PBR^{-1}B^TP = 0$

Similarly, for a discrete linear time-invariant system (assuming that the system is fully controllable), let the state equation be

$$X(k+1) = AX(k) + BU(k) \quad (18)$$

The

performance

indicators of the

secondary type

are

where Q is a symmetric positive definite constant array or symmetric semi-

positive definite constant array and R is a symmetric positive definite constant

array. The problem now is to design a state feedback controller $U(k) = -LX(k)$ that

is minimized. Above only the corresponding conclusions are given here:

$$L = (R + B^TPB)^{-1}B^TPA \quad (20)$$

where P is the solution to the **Riccati** equation $Q - P + A^TPA - A^TPB(R + B^TPB)^{-1}B^TPA = 0$

有了上述对 LQR 控制器定性的说明后，接下来以自行车的平衡过程为实例说明 LQR 控制器的设计。由自行车的动力学模型可知，系统的状态方程为

$$\begin{pmatrix} \dot{\varphi} \\ \ddot{\varphi} \\ \dot{\delta} \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 \\ \frac{g}{h} & 0 & -\frac{v^2}{hw} \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} \varphi \\ \dot{\varphi} \\ \delta \end{pmatrix} + \begin{pmatrix} 0 \\ -\frac{bv}{hw} \\ 1 \end{pmatrix} \delta \quad (13)$$

The

output

equation

is

$$y = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \varphi \\ \dot{\varphi} \\ \delta \end{pmatrix} \quad (14)$$

Usually, we do not calculate the state feedback matrix by hand, but borrow the relevant functions from **MATLAB** to do the calculation, the

```
clc
g = 9.8; % Acceleration of gravity
h = 0.088; % Height of the center of mass of the bicycle
v = 0.634; % Bicycle travel speed
w = 0.167; % Distance between front and rear wheels and
            % ground contact points
b = 0.055; % Distance between the point of impact of the
            % rear wheel and the projection of the center of
            % gravity on the ground
A_21 = g/h.
A_23 = -v^2/(w*h);
B_21 = -(b*v)/(w*h).
A = [0 1 0; A_21 0 A_23; 0 0 0];
b = [0; b_21; 1].
c = [1 0 0; 0 0 1].
```

relevant program is shown below:

```

D = 0;
Ts = 0.020; % Sampling interval
t = 0:Ts:4;
u = zeros(size(t));
[G,H] = c2d(A,B,Ts); % Transformation of continuous systems into discrete systems
x0 = [0.0873; 0; 0]; % Setting the initial state of the system
Tc = ctrb(G,H);
if (rank(Tc)==3)
    fprintf('This system is controllable! \n');
    Q = [300 0 0; 0 0 0; 0 0 0 300]; % Q matrix
    R = 1; % R matrix
    K = dlqr(G,H,Q,R); % Calculate state feedback coefficients (see calculated values for yourself)
    G2 = G-H*K;
    y = dlsim(G2,H,C,D,u,x0); subplot(2,1,1)
    plot(t,y(:,1),'b.-','LineWidth',1.5);
    xlabel('Time(s)');
    ylabel('\phi(rad)');
    grid on
    subplot(2,1,2)
    plot(t,y(:,2),'b.-','LineWidth',1.5);
    xlabel('Time(s)');
    ylabel('\delta(rad)');
    grid on
end

```

The result after running is shown below:

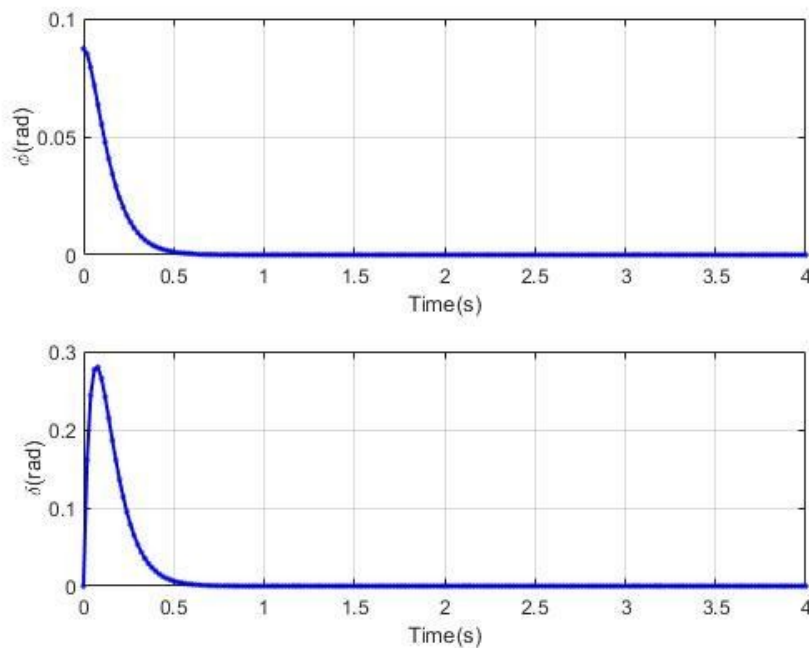


Fig. 2-2 LQR simulation results for discrete-time system

The corresponding C code is shown below:

```
float k1=-92.2973, k2=-8.6746, k3=10.5355;    //LQR feedback coefficient
...
delta_dot = -( k1*(phi-balance_point) + k2*phi_dot + k3*delta );
...
```

2.2 PID Controller

Since the steering speed of the fork at the output of the above LQR controller $\dot{\phi}$ cannot be obtained directly by controlling the servo (effective modeling of the servo is not possible) and since the servo is an open-loop system for the user, it is considered to incorporate a PID controller at the output of the LQR controller.

In our problem, its control rate can be easily obtained:

$$PWM[n] = K_p \cdot e[n] + K_d \cdot (e[n] - e[n-1]) + K_i \cdot \sum_{i=0}^n e[n] \quad (21)$$

where $PWM[n]$ corresponds to the magnitude of the steering angle of the servo and $e[n]$ denotes the deviation from the steering ϕ of the forks

The output of the LQR controller is used as the input to the PID controller because the output of the LQR controller is larger when the bike is leaning, so the output of the LQR controller is used as the input to the PID controller to make the output of the LQR controller go to 0 as fast as possible, and the deviation of the corresponding system at this time is also 0. Therefore, the equation (21) can be changed to

$$PWM[n] = K_p \cdot \delta[n] + K_d \cdot (\delta[n] - \delta[n-1]) + K_i \cdot \sum_{i=0}^n \delta[n] \quad (22)$$

Further on there are

$$PWM[n-1] = K_p \cdot \delta[n-1] + K_d \cdot (\delta[n-1] - \delta[n-2]) + K_i \cdot \sum_{i=0}^{n+1} \delta[n-1] \quad (23)$$

Equation (23) minus (22) yields the expression for incremental PID control

$$PWM[n] = PWM[n-1] + K_i \cdot \delta[n] + K_p \cdot (\delta[n] - \delta[n-1]) + K_d \cdot (\delta[n] - 2\delta[n-1] + \delta[n-2]) \quad (24)$$

The corresponding C code is shown below:

```
int ki=4, kp=16, kd=4; //PID parameter (integer for easy debugging after 100x
magnification)
...
u = ki*delta_dot + kp*(delta_dot-last_delta_dot) +
kd*(delta_dot-2*last_delta_dot+last_delta_dot_2);
Servo += u/10000;
...
```

3. bibliography

- [1] Limebeer D, Sharp R S. Bicycles, motorcycles, and models[J]. IEEE control systems, 2006, 26(5):34-61.
- [2] Liu Yanzhu. On the stability of bicycles[J]. Mechanics and Practice, 2012, 34(2):4.

4. appendice

4.1 About the steering structure

Regarding the steering structure of a bicycle, let's idealize it as a trapezoidal structure as shown below

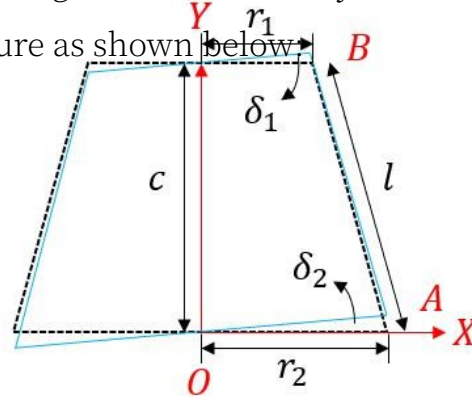


Figure 4-1 Steering Structure of Self-Balancing Bicycle

At rest, there are $(r_2, 0) = (r_1, 0)$; after rotation, there are $A' = (r_2 \cos \delta_2, r_2 \sin \delta_2)$, $B' = (r_1 \cos \delta_1, c + r_1 \sin \delta_1)$. Since the distance between and remains constant during the rotation, we have

$$\overline{AB}^2 = \begin{cases} (r_1 - r_2)^2 + c^2 = l^2 \\ (r_1 \cos \delta_1 - r_2 \cos \delta_2)^2 + (c + r_1 \sin \delta_1 - r_2 \sin \delta_2)^2 = l^2 \end{cases} \quad (25)$$

carry

out

$$2r_1r_2(1 - \cos \delta_1 \cos \delta_2) + 2c(r_1 \sin \delta_1 - r_2 \sin \delta_2) - 2r_1r_2 \sin \delta_1 \sin \delta_2 = 0 \quad (26)$$

When the rotation angle between the servo and the bicycle handlebar is small, it can be considered that $\delta_1 \rightarrow 0$, $\delta_2 \rightarrow 0$. Then, the

The equation(26) can be reduced to

$$\delta_1 = \frac{r_2}{r_1} \delta_2 \quad (27)$$

Therefore, it can be approximated that the bicycle handlebar angle is proportional to the servo angle.

4.2 About the system block diagram

The system block diagram of the above control system can be drawn as follows:

